

WGU D076 Complete Study Guide

WGU D076 — Finance Skills for Managers

COMPLETE STUDY GUIDE (Pass-Focused)

Sources: Your unit tests & module quizzes (with correct answers) + instructor cohort transcripts (Units 3, 4, Module 9 Forecasting I/II, Unit 6)

How to use: (1) Memorize the **OA Decision Cheat Sheet** first, (2) drill **Key Concepts**, (3) quiz yourself with the **Practice Q&A** banks.

OA Decision Cheat Sheet (Memorize First)

Capital budgeting (Unit 6) — which method?

Situation	Use
Only one project / maximize \$ value / mutually exclusive	NPV (highest NPV wins)
Limited capital; rank several projects	Highest PI first until budget used
Communicate return as a % / easy story	IRR (if conventional cash flows)
Unconventional CFs (multiple sign changes)	NPV (multiple IRRs problem)
Accept NPV	$NPV > 0$
Accept IRR	$IRR > \text{hurdle} / \text{cost of capital}$
Accept PI	$PI > 1$ ($PI = 1 \rightarrow IRR = \text{cost of capital}$; $NPV \approx 0$)
Answers shown as \$	Likely NPV
Answers shown as %	Likely IRR

Excel TVM (Unit 3 — most tested unit per instructors)

Need	Function / tip
Money needed now for future withdrawals	PV
Grow money to a future amount	FV
Uneven CFs → PV in one step	NPV
Uneven CFs → rate of return	IRR (cash flows only)
Ordinary annuity (end of period)	TYPE 0
Annuity due (beginning / today)	TYPE 1
Argument order	rate, nper, pmt, pv, type
Money leaving your hands	usually negative PMT

Forecasting (Unit 5 / Module 9)

Idea	Rule
Spontaneous accounts	Vary with sales (COGS, cash, AR, inventory, AP)
Discretionary	Management choice (notes payable, LT debt, common stock)
DFN	Proj Assets – Proj Liab – Proj Equity (after pro formas)
Negative DFN	Enough financing already
↓ DFN	↑ plowback/retention, ↑ net margin; check fixed assets first
SGR	Grow without new equity ; ↑ SGR by ↓ dividends / ↑ retention
Fixed assets	Lumpy — buy whole factory, not a fraction

Ratios & DuPont (Unit 4)

Idea	Rule
Suppliers / ST creditors care about	Liquidity
Quick vs current	Quick excludes inventory
High inventory turnover vs peers	May mean not enough inventory
ACP too high vs industry	Tighten credit standards
Debt ratio 40%	40% of assets financed by debt
Operating margin	Compare firms with different capital structures
Net margin 7%	7¢ per \$1 sales for equity holders
M/B > 1	Growth expected; M/B < 1 = value stock
DuPont ROE	Net Margin × Asset Turnover × Leverage
↑ ROE	↑ debt financing and/or ↑ net margin
Trend	Same firm over time
Cross-sectional / benchmark	vs peers / industry
Progress	vs a goal

Cash flows to include / exclude (Unit 6)

Include	Exclude
Incremental CFs	Allocated overhead
Incidental CFs	Sunk costs (prior consulting, prior market study)
Cannibalization of your own products	—
Opportunity cost (e.g., forgone land sale offer)	Already-spent purchase if offer is the true opp. cost focus

Indicators / ethics / agency (Unit 2)

Item	Answer
Leading	Yield curve (before economy)
Coincident	Personal income, GDP (during)
Lagging	CPI, unemployment (after)
Inverted yield curve	Possible downturn
Firm goal	Maximize owner wealth
Personal goal	Maximize utility
Ethical / Moral / Legal	Standards of conduct / personal beliefs / laws
Agency fix	Align via ownership / executive compensation

Instructor Cohort Highlights (Your Transcripts)

Unit 3 — Fundamental Financial Principles (instructor)

- **Unit 3 is the most tested unit on the exam.**
- Covers interest/required return/inflation, TVM Excel, and risk & return.
- Excel on PA/OA uses **compound interest** concepts.
- Simple interest = straight line ($P \times R \times T$); compound = exponential (interest on interest).
- Practice every Excel problem type before the PA/OA.

Unit 4 — Financial Ratios Review (instructor)

- **Trend** = same firm over time; **Progress** = toward a **goal**; **Cross-sectional** = vs peers; benchmarking is a form of cross-sectional.
- Banks & **suppliers** care about **liquidity**.
- **Quick ratio** (acid test) excludes inventory — always $\leq$ current ratio.
- **DuPont** focuses you on what drives ROE: profitability, asset efficiency, leverage.

Module 9 — Financial Forecasting I & II (instructor)

- Spontaneous accounts **react to sales**.
- DFN = projected assets – projected liabilities – projected equity.
- **Lumpy assets** = purchased in large non-divisible chunks (new machine/factory).
- Higher retention & higher ROE → higher SGR; firms dislike issuing new shares (dilution/price hit).
- Higher earnings retained → more internal equity → **decreases DFN**.
- Interest expense on pro formas can be independent of sales/operations.

Unit 6 Cohort (instructor)

- OA is about **application**, not just terminology.
- **One project / choose one** → start with **NPV** (exact \$ value added).
- NPV advantages: all CFs, TVM, cost of capital/risk built in. Disadvantages: hard cost of capital; bad for different sizes.
- IRR easy as %; don't use for mutually exclusive or multiple-IRR projects; prefer NPV when conflict.
- PI = benefit/cost ratio; best when **budget constraints**; prioritize highest PI.
- NPV = how much value; PI = how efficiently value is created.

Unit 2 — Foundations of Finance

Key Concepts to Remember

Areas of Finance

Topic	What it means	Remember
Business finance	Sources of funding, capital structure of corporations; seeks to increase firm value to owners	Uses/sources of funding to increase firm value
Investments	Decide which assets to invest in to create wealth / earn positive returns in the future	Allocate money for future wealth
Financial institutions	Accept deposits, offer investment products, provide loans, broker transactions; primary role = conduct financial transactions (investments, loans, deposits) to circulate money	Circulate money via transactions
Asset pricing	Determining the value/price of financial assets	Valuation focus

Finance vs Accounting (MUST KNOW)

	Finance	Accounting
Time focus	Future -looking	Generally backward -looking
Role	Manage/allocate capital: investing, forecasting, budgeting, saving, lending, borrowing	Historical financial records/data

Goals: Firm vs Personal Finance

Context	Goal / aim
Firm / financial manager	Maximize owner wealth (NOT manager's utility; NOT just minimize costs)
Personal financial goals	Maximize individual utility / maximize satisfaction from products & services obtained
Any investment decision — main question	Do the benefits outweigh the costs? (benefit $\geq$ initial cost)

Personal finance tools / actions: - **Budgeting** = tool to understand overall personal cash flows (income & expenses) - Emergency stash alternative = **savings account** (readily withdrawable + some interest) — NOT high-risk stocks, long-term bonds, or a nicer car - Affordable mortgage for a house = **financing** (finance goals within your means) - Already decided to buy; choosing how to pay = **financing a goal** - **Financial planner** = works with **individuals** to achieve financial goals

Three Types of Financial Manager Decisions

Decision type	What it is	Test cues
Investment decision	Assess costs/benefits of potential projects	R&D worth it?; assessing costs/benefits of projects
Financing decision	Where to get funds / capital structure	Issue stocks/bonds; obtain a loan to buy equipment; Hannah raises \$5M after approval
Working capital management	Day-to-day cash flows	Inventory, receivables, payables

Financial Markets & Regulation

Item	Definition / role
Purposes of financial markets	Provide liquidity and determine prices
Primary market	First-time issuance (IPO / new stock to raise capital)
Secondary market	Trading after initial issuance
Money market	Institutions borrow/lend short term ; highly liquid; typically ≤ 1 year
SEC	Protect investors; fair/orderly/efficient markets; capital formation
Central bank / Fed	Controls money circulating; adjusts rates to regulate inflation and unemployment

Types of Financial Institutions

Type	Role / definition	Test cue
Depository	Accepts & pays interest on deposits; extends loans	Banks, credit unions
Contractual	Insurance companies	Insurance = contractual
Insurance (how claims are paid)	Invest premiums in stocks/bonds; use returns to pay claims	Not “withdraw from premium accounts”
Investment institution	Provides individuals & firms access to financial markets	Mutual funds
Pension fund	Retirement through employers	Employer retirement
Private equity	Invest in firms not on public exchanges ; buyout troubled companies → improve → sell/IPO	Non-public / troubled buyouts
Investment bank	Complex deals like mergers	M&A help
Central bank	Nation’s money supply / economy health	Fed

Economic Indicators — Leading, Lagging, Coincident (MUST KNOW)

Type	When it moves	Examples from YOUR tests	Extra tested facts
Leading	Before economy changes	Yield curve	Inverted curve → may signal economic downturn (low/negative future growth)
Coincident	During / with the economy	Personal income	Useful: analyzed during shifts to show current state of economy (big picture). Note: GDP & personal income are coincident — they do not predict the future
Lagging	After economy changes	CPI, unemployment rate	Confirm what already happened

Memory: Leading = ahead | Coincident = with | Lagging = behind

Ethics Terms (Distinguish Carefully)

Term	Meaning	Test note
Ethical	Based on accepted standards of conduct	Society/industry accepted standards
Moral	Person's beliefs about right/wrong, good/bad, just/unjust	Personal sense of right/wrong
Legal	In accordance with laws/rules of an authority	Legal error → government penalty/repercussion
Standard	Measure, norm, or model for comparison	

Moral vs ethical trap (Lucas / Bullzai): - Practice is legal + commonly accepted in industry → **not** legal issue, **not** ethical issue - But advisor knows it's not in clients' best interest → **moral** dilemma (his own sense of right/wrong)

Nora (investment manager) ethical dilemma: - Considering a risky client investment **just to meet her monthly goal** = ethical dilemma (duty to client vs personal goal)

Other ethics applications: - Maximize shareholder value **unethically** → often **decreases** shareholder value (costly results) - Moving production overseas: **unethical** = saving money by paying **inadequate wages** to overseas workers - Advisor (Jack): may recommend firm product if you **explain performance vs market** and don't hide alternatives

Steps to Solve an Ethical Dilemma

1. Identify and define the problem
2. Consider alternative courses of action ← **2nd step**
3. **Consider all stakeholders involved** ← **3rd step** (confirmed)
4. Move forward with the chosen course of action

Agency Problems & Bondholders

Concept	Remember
Agency problem	Managers follow own interests instead of owners'
Agency examples	Fake accounting + sell inflated stock; manager buys company car as company expense (luxury, doesn't help shareholders)
Why manipulate accounting?	Make performance look good → bonuses / stock benefits
Mitigate agency costs	Align managers' interests with shareholders' (usually ownership shares / executive compensation)
Bondholders want	Safe projects with higher chance of being paid back
Why strict bond contracts/covenants?	Bondholders primarily want to be paid back ; deter risky projects
Bank on risky loan	Set a strict covenant company can't easily achieve

Practice Questions — Unit 2 Form A

1. Assets for future wealth → **Investments**
2. Main goal of firm → **Maximize owner wealth**
3. R&D evaluation → **Investment decision**
4. After initial issuance → **Secondary market**
5. Insurance company type → **Contractual**
6. Company retirement funds → **Pension fund**
7. Personal income → **Coincident**
8. Standards of conduct → **Ethical**
9. 2nd ethical step → **Consider alternative courses of action**
10. Reduce agency via corporate control → **Executive compensation**

Practice Questions — Unit 2 Form B

1. Institution services → **Deposits, investment products, loans, brokering transactions**
 2. Personal goals objective → **Maximize individual utility**
 3. Issuing stocks/bonds → **Financing decisions**
 4. Why finance definition matters personally → **Compare costs and benefits**
 5. First-time issue → **Primary market**
 6. Mutual fund → **Investment institution**
 7. Administer retirement funds → **Pension funds**
 8. CPI → **Lagging**
 9. Ethical means → **Accepted standards of conduct**
 10. Fake accounting + sell stock → **Agency problem due to conflicting interests**
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Practice Questions — Unit 2 Module Assessment Section 1 (earlier sets)

Personal / manager tasks: Investments; Hannah \$5M → financing; Spain trip goal → maximize individual utility; Financial planner; Omar car pay → financing a goal

Markets / institutions / indicators: SEC → protect investors; Access to markets → investment institutions; Troubled buyouts → private equity; Yield curve → leading; Inverted curve → downturn

Ethics / agency: Legal = laws of authority; Jack → recommend + explain vs market; Manipulate accounting → look good; Agency = managers' interests ≠ owners'; Bank → strict covenant

Practice Questions — Unit 2 Module Assessment Section 1 (new batch)

Business finance / finance vs accounting / investments

1. Sources of funding + capital structure + increase firm value → **Business finance**
2. Primary difference finance vs accounting → **Finance focuses on the future; accounting is generally backward-looking**
3. Subspecialty deciding which assets create wealth/positive returns → **Investments**

Goals

1. Primary goal of financial manager → **Maximize owner wealth**
2. Main question for any investment → **Do the benefits of this investment outweigh the costs?**
3. Primary aim of personal finance goals → **Maximize satisfaction from products purchased and services obtained** (= utility)

Manager tasks / careers

1. Obtain loan to buy equipment for a project → **Making financing decisions**
2. Invest in firms not on public exchanges → **Private equity**
3. Assess costs and benefits of potential projects → **Making investment decisions**

Personal cash tools

1. Tool for overall personal cash flows → **Budgeting**
2. Reasonable alternative to emergency cash stash → **Investing in a savings account**
3. Affordable mortgage to buy a house → **Financing**

Markets

1. Purposes of financial markets → **Provide liquidity and determine prices**
2. Stocks/bonds after initial issuance → **Secondary market**
3. Institutions borrow/lend short term → **Money market**
4. IPO / issue public stock first time → **Primary market**

Institutions (roles)

1. Primary role of financial institutions → **Conduct financial transactions such as investments, loans, and deposits**
2. Depository institution → **Accepts and pays interest on deposits and extends loans**
3. Controls money circulating / economy healthy → **Central bank**
4. How insurance pays claims → **Use returns from stocks and bonds**
5. Retirement through employers → **Pension fund**
6. Help large corps merge → **Investment bank**

Indicators / Fed

1. Unemployment rate → **Lagging**
2. Fed adjusts rates to → **Regulate inflation and unemployment**
3. Inverted yield curve signals → **Economic downturn**
4. Coincident indicators useful because → **Analyzed during economic shifts to provide information about the current state of the economy**

Ethics distinctions & dilemmas

1. Beliefs about right/wrong → **Moral**
2. Ethical action characterized by → **Accepted standards of conduct**
3. Lucas/Bullzai (legal + industry-accepted, but not client-best) → **Moral dilemma**
4. Error with government penalty → **Legal**
5. Nora investing clients in risk just for monthly goal → ethical dilemma because **considering risky asset just to meet her monthly goal**
6. Maximize shareholder value unethically → **Often leads to decreased shareholder value**
7. Unethical overseas move → **Paying inadequate wages to workers overseas**

Bondholders / agency / ethical steps

1. Strict bond contracts to deter risk → bondholders want to **be paid back**
2. Bondholders interested in → **Safe projects with higher chance of sufficient compensation**
3. Agency example → **Manager purchases a company car as company expense**
4. Mitigate agency costs → **Aligning managers' interests with shareholders' interests**
5. **Third** step in ethical dilemma → **Consider all stakeholders involved**

Quick Memory Hooks (Unit 2)

- **Business finance** = funding + capital structure + firm value
- Finance = **future**; accounting = **past**
- Firm goal = **owner wealth**; personal = **utility/satisfaction**
- Investment test = **benefits $\geq$ costs**
- Assess projects = **investment**; get a loan/raise funds = **financing**
- Budget = cash flows; emergency cash alt = **savings account**; mortgage = **financing**

- Markets purpose = **liquidity + prices**; short-term = **money market**
- IPO = **primary**; later trading = **secondary**
- Institutions' role = **transactions** (investments/loans/deposits)
- Depository = deposits + loans; insurance pays from **investment returns**; pension = employer retirement; PE = private/non-public; IB = mergers; central bank = money supply
- SEC = **protect investors**; Fed = **inflation & unemployment**
- Leading = yield curve; Coincident = personal income (+ current-state use); Lagging = CPI + **unemployment**
- Inverted curve → **downturn**
- Ethical = standards; Moral = personal beliefs; Legal = law + gov penalty
- Industry-accepted but wrong for client → **moral** (Lucas); risk for monthly goal → **ethical** (Nora)
- Unethical “max value” → often **hurts** value; inadequate overseas wages = unethical
- Agency = own interests / company car luxury; fix = **align** via ownership pay
- Bondholders = **safe projects** + paid back; banks use **strict covenants**
- Ethical steps: 1 problem → 2 alternatives → 3 **stakeholders** → 4 act

Unit 3 — Interest Rates, Required Return & Inflation

(OA-heavy area per Reddit: interest rate / required rate / inflation often ~30%)

Key Concepts to Remember

Interest Rate — Definition & Purpose

Concept	Remember
Interest rate	The percentage of the principal that a lender charges a borrower for the use of assets
How it is expressed	As a percentage (not a dollar amount, ratio, or probability)
Main purpose of charging interest	Borrowers pay to use assets/funds that belong to another entity to accomplish their own goals
Other names for interest rate (perspective)	Discount rate, required rate, cost of capital

Required Rate of Return — Components

Required rate of return is composed of: 1. **Opportunity cost** 2. **Risk** 3. **Inflation**

Idea	Remember
Long-term investments need higher required return	Greater risk (harder to ensure return over longer time) + higher opportunity cost (money tied up longer)
Opportunity cost example	Inherit \$25k → put in savings → opportunity cost could be buying a brand new \$25k car (what you give up)
Not opportunity cost	Bank fees; emergency access; earning interest on the savings
Inflation example (Ahmed)	Saved exactly \$15k for a car; car now costs \$16,562 → he forgot inflation

Inflation

Cause / idea	Remember
Increased demand	Demand ↑ → insufficient supply → prices go up until demand = supply again
Built-in inflation & adaptive expectations	Prices ↑ → workers demand higher wages to keep living standard → pushes prices even higher
What happens to prices with inflation?	Prices rise (from demand ↑, cost ↑, adaptive expectations)

Risk Premium, Nominal Rate, Risk-Free Rate, APR

Term	Meaning
Risk premium	Compensation for risk given to investors
Nominal rate	Rate at which invested money grows for a certain period; interest rate most often used in daily life; includes inflation → represents amount of money you'll have in the future
Risk-free rate	Return on an investment with no risk ; indicates inflation and opportunity cost (not risk)
Real rate	Reflects future purchasing power / rate adjusted to remove effects of increased prices (contrast with nominal)
Annual percentage rate (APR)	Interest rate expressed on an annual basis; annual interest rate charged for borrowing or earned through investment
Hurdle rate	Another name for required rate of return — the minimum rate a firm must surpass to accept a project
Inflation rate	Rate at which the average price level of a basket of goods and services in an economy increases

Sophia stock example (application): - Inflation 2–3%, Treasury ~2%, stock has significant risk - Expect return **higher than inflation** to cover opportunity costs and risks - Inflation reduces nominal returns → need higher return for risk + opportunity cost - Do **not** accept less than Treasuries when risk is higher; company doesn't set her required return; still consider opportunity costs

Module summary hooks: - Interest rate = % of principal lender charges borrower

- Also called discount rate / required rate / cost of capital / **hurdle rate**
 - Composed of opportunity cost, risk, inflation
 - Inflation from increased demand, rising costs, built-in inflation
 - Nominal includes inflation (future money amount); real = purchasing power
 - Risk-free rate = inflation + opportunity cost
 - APR = interest rate on an **annual** basis
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Practice Questions — Unit 3 Assessment Section 1

Interest rate basics

1. Term for % of principal lender charges borrower for use of assets → **Interest rate**
2. How is interest rate expressed? → **As a percentage**
3. Main purpose of charging interest? → **Allows borrowers to pay to use the assets of another entity to accomplish their own goals**

Required return / opportunity cost / inflation

1. Component of required rate of return? → **Opportunity cost** (also risk & inflation make up the required rate)
2. Why long-term investment requires higher rate of return? → **Greater risk and higher opportunity cost**
3. Inherit \$25k in savings — opportunity cost? → **Buying a brand new car worth \$25,000**
4. Ahmed saved \$15k for a car now priced \$16,562 — which component did he forget? → **Inflation**

Inflation causes / effects

1. Why is built-in inflation linked to adaptive expectations? → **Workers want higher wages as prices increase, which pushes prices even higher**
2. Why does increased demand cause inflation? → **Insufficient supply → prices go up until demand equals supply again**
3. What happens to prices with inflation? → **Prices rise**

Risk premium / nominal / risk-free

1. Compensation for risk given to investors → **Risk premium**
2. Rate at which invested money grows for a certain period → **Nominal rate**
3. Component of interest rate that indicates inflation and opportunity cost → **Risk-free rate**

Unit 3 Module 1 quiz (APR / hurdle / inflation rate / Sophia)

1. Interest rate expressed on an annual basis → **Annual percentage rate (APR)**
2. Why is required rate of return also called the hurdle rate? → **It is the minimum rate that a firm must surpass to accept a project**

3. What is the inflation rate? → **Rate at which the average price level of a basket of goods and services in an economy increases**
 4. What does the risk-free rate indicate? → **Inflation and opportunity cost**
 5. Sophia: risky popular stock, inflation 2–3%, Treasuries ~2% — what return to expect? → **Based on inflation, expect a return higher than this for the associated risk** (cover opportunity costs + risks; inflation reduces nominal returns)
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Quick Memory Hooks (Unit 3 so far)

- Interest rate = **% of principal**; expressed as **percentage**; annual form = **APR**
 - Purpose = pay to use someone else's assets/funds
 - Same idea aka: **discount rate / required rate / cost of capital / hurdle rate**
 - Hurdle = **minimum** return project must beat to be accepted
 - Required return = **opportunity cost + risk + inflation**
 - Longer term → **more risk + more opportunity cost** → higher required return
 - Opportunity cost = what you **give up** (e.g., the car you don't buy)
 - Price rose while saving → forgot **inflation**
 - Inflation rate = ↑ in average price of a **basket of goods/services**
 - Demand inflation = shortage → prices ↑; built-in = wages chase prices
 - Inflation → **prices rise**
 - Risk compensation = **risk premium**
 - Growth of money in daily life = **nominal** (includes inflation)
 - Risk-free rate = **inflation + opportunity cost** (no risk)
 - Nominal = future **money amount**; Real = **purchasing power**
 - Risky stock + inflation → expect return **above inflation** for risk/opportunity cost
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Unit 3 — Time Value of Money (TVM)

Key Concepts to Remember

TVM Core Idea

Concept	Remember
Time value of money	Money available today is worth more than the same amount in the future
Why / what it considers	Amount of cash flows at different times with a certain interest rate
Present value (PV) & Future value (FV)	Relative to other cash flows

Compounding vs Discounting

Process	Direction	Meaning
Compounding	PV $\rightarrow$ FV	Finding a future value given a present value
Discounting	FV $\rightarrow$ PV	Finding a present value given a future value

Annuities

Type	Definition / cue
Annuity (general)	Equally spaced cash flows of equal amounts
Ordinary annuity	Series of equal payments at the end of consecutive periods over a fixed length of time
Annuity due	Equal payments at the beginning of periods (contrast with ordinary)
Perpetuity	Annuity that continues forever (not a fixed length)

Ordinary annuity keys (tested): “at the **end** of each period” + “**fixed** period” + “**equal** payments”

TVM Example (quiz)

- Invest \$10,000 today + \$5,000/year for 5 years at 4% $\rightarrow$ can withdraw **\$39,248.14** in 5 years
- That \$39,248.14 = **Future value** (worth of relative past cash flows; relative future to other cash flows)

Excel / TVM functions (from lesson + quizzes)

Idea	Remember
Excel TVM functions	Use function + cell references for inputs so changes update automatically (don't hard-code numbers if inputs are in cells)
TYPE input	Cash flow timing: 0 = END (ordinary annuity); 1 = BEGIN (annuity due)
Annuity due PV example	\$5,000/year for 20 years, 3% discount → TYPE = 1
FV function argument order	FV(rate, nper, pmt, pv, type) — reference cells in that order
Ordinary annuity FV example	Save \$1,000/year for 10 years at 3% → =FV(C2, C3, C4, C5, C6) style (rate, nper, pmt, pv, type); TYPE typically 0 for ordinary
When to use PV	Finding lump sum needed now to support future withdrawals / relative past value of future cash flows (e.g., withdraw \$10k/year for 5 years at 3%); also FV/PV of a single sum given an interest rate
When to use FV	Finding future worth of current/past cash flows (single sum or annuity)
Also learn	PMT, RATE, NPER Excel functions
When to use NPV	Present value of uneven cash flows in one step
When to use IRR	Rate of return on uneven cash flows; references cash flows only (not the discount rate)
IRR example	To find interest rate of a cash-flow series → =IRR(B3:E3) (range of cash flows only)
Other functions mentioned	NPER, RATE (not the right pick for “lump sum needed now”)

Compounding interest: Interest is calculated on the **principal + earned interest** (interest on interest).

Sign/timing tip (verify in Excel lessons): payments “starting today” / beginning → TYPE 1; end of period → TYPE 0; money leaving usually negative for PMT.

TVM in Business & Personal Decisions

Concept	Remember
TVM influences	Both business and personal decision-making
Break-even vs dollars received	Because of TVM, the break-even amount for an investment is not comparable to the actual dollar amount you receive over the investment period (must adjust for time)
Why TVM matters in decisions	Benefits received at different times are comparable only when you consider TVM — find today's value of future cash flows to compare costs and benefits
Loan / mortgage length	Longer period → higher interest charged over life of loan → more total interest paid (even at the same rate)

Mortgage example (tested): \$250k house; 20-year vs 30-year mortgage, both at 3.5% → **30-year** requires more **total interest** (same rate, longer life = more interest paid).

Practice Questions — Unit 3 TVM Assessment Section 1

1. Series of equal payments at the **end** of consecutive periods over a fixed length of time → **Ordinary annuity**
2. \$10k today + \$5k/year for 5 years at 4% → withdraw \$39,248.14 in 5 years. What is \$39,248.14? → **Future value**
3. Concept that a dollar today is worth more than a dollar in the future → **Time value of money**

TVM decision-making quiz

1. \$250k house; 20-year vs 30-year mortgage, both 3.5% — which costs more total interest? → **A 30-year mortgage**
2. Why does TVM play an important role in financial decision-making? → **Benefits of investments received at different times are comparable only when you consider the time value of money**

Excel TVM functions quiz

1. PV of annuity due \$5,000/year for 20 years at 3% — what is TYPE? → **1** (BEGIN / annuity due)
2. FV of ordinary annuity; inputs in C2:C6 — correct formula? → **=FV(C2,C3,C4,C5,C6)** (rate, nper, pmt, pv, type cell refs)
3. Lump sum needed now to withdraw \$10,000/year for 5 years at 3% — which Excel function? → **PV function**

Compounding / uneven cash flows (NPV & IRR) quiz

1. Present value of **uneven** cash flows in one step → **NPV**
 2. Find interest rate of a cash-flow series (discount rate shown but not used in formula) → **=IRR(B3:E3)** (IRR of cash flows only)
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Quick Memory Hooks (Unit 3 TVM so far)

- TVM = **today's dollar** > **future dollar**
- **Compounding** = PV → FV; **Discounting** = FV → PV
- Compounding interest = interest on **principal** + **earned interest**
- PV/FV are **relative** to other cash flows
- Annuity = equal, equally spaced cash flows
- **Ordinary** = end of period (TYPE **0**); **Annuity due** = beginning (TYPE **1**); **Perpetuity** = forever
- Lump sum today + payments growing to a later amount → that later amount is **FV**
- TVM lets you compare costs/benefits across **different times** (today's value of future CFs)
- Longer loan life → **more total interest** (even at same rate)

- Break-even \$ $\neq$ raw dollars received over time without TVM
- Excel equal CFs: PV, FV, PMT, RATE, NPER
- Excel uneven CFs: **NPV** = PV in one step; **IRR** = rate of return (cash flows only)
- Need money **now** for future withdrawals $\rightarrow$ **PV**; grow money to later date $\rightarrow$ **FV**

Unit 3 — Returns (NEW from this batch)

Key Concepts

Term	Definition
Return	Money gained or lost on an investment over some period of time
Holding period return	Return over the entire period an investor owns a financial security
Expected return	Hypothesized estimate of future returns under different scenarios based on expectational data / probability of scenarios
Real rate (purchasing power)	Roughly nominal return – inflation — use to compare purchasing power across years

Purchasing power example: 1980: inflation 5%, return 15% $\rightarrow$ real $\approx$ 10%. 2010: inflation 5%, return 12% $\rightarrow$ real $\approx$ 7%. Stockholders gained greater purchasing power in **1980** because the **real rate** was higher.

Practice — Returns

1. What is a return? $\rightarrow$ **Gain or loss on an investment over some period of time**
2. What is an expected return? $\rightarrow$ **Hypothesized estimate of future returns under different scenarios based on expectational data**
3. 1980 vs 2010 purchasing power (same 5% inflation; returns 15% vs 12%) $\rightarrow$ **1980 because the real rate was higher**
4. What makes expected return subjective/different? $\rightarrow$ **Based on expectational data and the probability of different scenarios occurring**

Risk (Market / Firm-Specific / Reduction)

Key Concepts

Concept	Remember
Risk (in finance)	Possibility that the realized/actual return will differ from the expected return
Systematic / market risk	Economy-wide; cannot be diversified away
Firm-specific / idiosyncratic risk	Company problems (lawsuits, labor, management, etc.); can be diversified away
Interest rate risk	Market risk where bond value is affected by changes in interest rates
Default risk	Firm-specific risk from probability of loss if borrower fails to repay a contractual obligation
Price risk — operating risk	Depends on the effect of the firm's operating decisions on its operating costs

Risk reduction techniques

Technique	Meaning / example
Diversification	“Spreading” money over many different assets; reduces firm-specific risk (only up to a point — market risk remains)
Risk separation	Dispersing assets geographically instead of concentrating in one location (differs from diversification)
Risk transfer	e.g., buying home insurance — risk transferred from policyholder to insurer
Risk retention	Keep risk when cost of pursuing the activity is believed less than the alternative

Risk–return & time

Idea	Remember
Systematic risk differs by firm	Utility companies → low systematic risk (move with market in a diminished way); luxury often higher
Higher return	Typically higher risk / greater uncertainty
Time diversification	Stock investments are more risky over a shorter period than over a longer period
CAPM idea (lesson)	Trade-off between risk and return; higher systematic risk → higher expected return

Practice — Risk

1. Risk in finance → **Actual/realized return may differ from expected return**
2. Market vs firm-specific → **Market cannot be diversified away; firm-specific can**
3. Firm-specific risk → Problems from lawsuits, labor, management, etc.
4. Interest rate risk → **Market risk**; bond value affected by rate changes
5. Risk reduced by variety of assets in portfolio → **Firm-specific risk**
6. Risk separation vs diversification → Separation = disperse assets **geographically**
7. Risk transfer example → **Buying home insurance**
8. Why retain risk? → Cost of activity believed **less than the alternative**
9. Utilities' systematic risk → **Low** (diminished co-movement with market)
10. Time & stocks → **More risky short-term than long-term**
11. Higher return tends to mean → **Higher risk**
12. Competitor hit + wage cuts → labor strike → **Idiosyncratic / firm-specific risk**
13. Price risk type correctly identified → **Operating risk** (operating decisions → operating costs)
14. Spreading money over many assets → **Diversification**
15. Default risk → **Firm-specific**; loss if borrower fails to repay

Ratio Analysis (OA-heavy)

Why ratios matter

Idea	Remember
Purpose	Compare companies of different sizes/strategies/risk; evaluate performance; know what to focus on
Flexibility	Not regulated — can create new ratios as needed (e.g., cash / short-term obligations when AR slow & inventory illiquid)
Standardization	Makes firms of different sizes comparable
Useful through comparisons	Trend analysis, cross-sectional analysis, progress measurement
Shareholders use	Help determine whether firm is maximizing shareholder wealth
Benchmarking	Compare firm to similar firms / peer group

Comparison methods

Method	Meaning
Trend analysis	Firm's performance over time
Cross-sectional analysis	Firm vs competitors / industry / peer group
Progress measurement	Compare current ratio to a goal and track progress toward it
Benchmarking	Performance relative to similar firms

Five major ratio types

Type	What it measures
Liquidity	Ability to meet short-term obligations (often without raising external capital). Suppliers care about these
Activity (efficiency / asset use)	How well assets generate sales or cash
Leverage / Financing	How firm is financed (debt vs equity); ability to pay interest & long-term obligations
Profitability	How well management maximizes shareholder wealth; performance & cost efficiency
Market	Evaluate current share price of public firm's stock

Key specific ratios

Ratio	Type / meaning
Current ratio	Liquidity — meet short-term obligations
Quick ratio	Liquidity — like current but excludes inventory (inventory least liquid; more stringent)
Inventory turnover	Activity — inventory management; high vs industry may mean not enough inventory (customers wait)
AR turnover	Activity — collecting receivables
Average collection period (ACP)	Days on average to collect AR. ACP 70 → takes 70 days to collect. If ACP 67 vs industry 40 → tighten credit standards
Total asset turnover (TAT)	Activity — sales per dollar of assets (higher = more efficient asset use)
Fixed asset turnover (FAT)	Activity — fixed asset efficiency
OIROI	Activity — operating income return on investment
Debt ratio	Leverage — % of assets financed by debt (40% → 40% of assets financed by debt)
Times interest earned (TIE)	Ability to pay interest / long-term obligations
Gross margin	Profitability — % of sales that become gross profit (after COGS); useful for production cost efficiency / profit per widget before other op costs
Operating margin	Profitability — pre-interest; useful to compare firms with different capital structures
Net margin	Profitability — % of revenue retained as profit (NI/Sales). 7% → 7 cents per \$1 sales remain for equity holders after all costs
Market-to-book (M/B)	Market — growth prospects; >1 expected to grow; <1 = value stock
Price-to-earnings (P/E)	Market — assess if stock/firm correctly valued

DuPont framework (MUST KNOW)

ROE = Net Margin × Total Asset Turnover × Leverage Multiplier

Also: **ROE** = **ROA** × **Leverage Multiplier** (ROA is a DuPont component path)

Idea	Remember
What DuPont shows	How profitability , asset management , and financing contribute to ROE
Component of ROE	Net margin (also TAT & leverage)
↑ ROE (all else constant)	Increasing debt financing (↑ leverage multiplier); ↑ net margin (cut costs); ↓ equity financing + ↑ net margin
Identical ROE, different financing	BigDog 40% debt vs SmallDog 100% equity → SmallDog has higher ROA (same ROE, smaller leverage multiplier)
Same PM & AT, higher ROE	Firm uses higher proportion of debt (larger leverage multiplier)
Improve ROE	Successfully cut production costs to boost net margin

Extra applications

- Paying off short-term loans more quickly → **liquidity ratio increasing**
- Different ratio types needed because they inform **different parts** of the firm
- Ratios useful across firm sizes because they provide **standardization**

Practice — Ratios (selected from quizzes)

1. Ratio use example → compare to peer group = **cross-sectional**
2. Why flexible? → **Not regulated; invent as needed**
3. Help shareholders? → Determine if maximizing **shareholder wealth**
4. Compare to similar firms → **Benchmarking**
5. Short-term obligations without external capital → **Liquidity ratios**
6. Why many ratio types? → Info about **different parts** of firm
7. Leverage ratios describe → **Proportions of equity and debt** financing assets
8. Paying ST loans faster → **Liquidity ratio increasing**
9. Current ratio type → **Liquidity**
10. Current vs quick → Quick **excludes inventory**
11. M/B = 1.2 → firm **expected to grow**
12. Net margin measures → % of revenue retained as **profit**
13. ROE component → **Net margin**
14. DuPont helps how? → Profits, assets, financing → ROE
15. ↑ ROE all else equal → **Increasing debt financing**
16. DuPont component → **Return on assets** ($ROE = ROA \times \text{leverage}$)
17. Improve ROE → **Cut costs / boost net margin**
18. Same ROE; one all-equity → all-equity has **higher ROA**
19. Actions that ↑ ROE → **Decrease equity financing + increase net margin**
20. Goal ratio 1.5 → 2 via cost cuts → **Progress measurement**
21. Market ratios purpose → Evaluate **current share price**
22. Profit per Widget (COGS only) → **Gross margin**
23. What shows what drives ROE changes → **DuPont framework**
24. Who uses assets more efficiently → Higher **asset turnover**
25. Performance over time → **Trend analysis**
26. vs competitors/industry → **Cross-sectional analysis**

- 27. Suppliers interested in → **Liquidity ratios**
- 28. Avg time to collect AR → **Average collection period**
- 29. Debt ratio 40% → **40% of assets financed by debt**
- 30. Operating margin useful for → Comparing profitability of firms with **different capital structures**
- 31. ACP of 70 → **70 days to collect AR**
- 32. High inventory turnover vs peers → **Not enough inventory**; customers wait
- 33. ROA vs ROE → ROE considers **capital structure**; ROA does not
- 34. Improve ROE (same leverage & AT) → **Reduce costs / improve profitability**
- 35. Ratios across sizes → **Standardization**
- 36. Create cash/ST obligations ratio → **Flexibility**
- 37. Activity = efficiency because → Measure asset use to generate sales/cash
- 38. How financed + pay interest/LT obligations → **Financing ratios**
- 39. Net margin 7% → 7¢ per \$1 sales for **equity holders**
- 40. ACP 67 vs industry 40 → **Tighten credit standards**
- 41. Current vs quick again → Inventory excluded from quick
- 42. M/B < 1 stock term → **Value stock**
- 43. Inventory turnover assesses → **Inventory management**
- 44. Same PM & AT, higher ROE → **Higher proportion of debt**

Unit 5 — Cash Budgeting & Financial Forecasting (OA-heavy)

A. Cash Budgeting (Short-term)

What / horizon / uses

Idea	Remember
Cash budget	Forecast of short-term events to see if firm has enough cash for operations
Time horizon	Usually 1 month to 1 year (not useful <1 month; not necessary >1 year)
Budgeting focus	Forecast future events for a specific short-term period (months up to a year)
3 main uses	(1) Forecast future financing needs , (2) basis for corrective action , (3) performance evaluation
Variances on cash budget	Can show certain managers/divisions not meeting targets (performance evaluation)

Six principles of personal budgeting

- 1. Know yourself
- 2. Understand key areas: **savings, income, expenses**
- 3. Develop savings, income, and expense strategies

4. Keep records

5. Use a method that meets your needs/objectives
6. Eliminate consumer debt and minimize long-term debt

Before personal budget, determine: income, expenses, and savings

Creating a cash budget (business) — order

1. Determine cash receipts
2. Estimate cash disbursements
3. Create the cash budget

Business cash budget should include: **cash receipts, cash disbursements, and borrowing**

Personal vs business budgeting steps

- Business: cash inflows, estimate outflows, create plan
- Personal: understand goals → track cash flows → develop plan → try plan → adjust
- Effective budget process ongoing: **Track** → **Monitor** → **Revise**

Cash receipts vs disbursements

	Business	Personal
Receipts	Cash sales + collected AR	Salary/wages; e.g., graduation gift
Disbursements	Materials, interest, taxes, rent, admin, selling expenses, etc.	Groceries, gas, insurance, etc.
Example receipt	Gift of \$100 from grandmother	
Example disbursement	Rent check paid/cashed this month; labor costs	
Not a disbursement this month	Inventory on credit paid next month	
Not a receipt	Collection timing aside; ski pass trade isn't cash income the same way	

Sales ≠ cash receipts: Sales include **cash + credit** sales; credit sales become receipts only when collected.
Example: \$100k sales but \$90k cash → firm did not make all sales on cash.

Fixed expenses / goals / efficiency

- Monthly mortgage = **fixed expense** (same amount each month)
- Better goal: “Put \$50 in savings each month for Christmas” → **specific and measurable**
- Make budget more efficient: compare budgeted vs actual cash flows, then **revise if needed**
- Implement budget changes: **prioritize and implement gradually, one by one**

Tracking, monitoring, revision

Term	Meaning
Tracking	Recording cash flows (envelope, spreadsheet, software)
Monitoring	Evaluate actual vs goals; find patterns; know when correction needed
Revision	After analysis; implement gradually
Why track?	Recognize where/how money is spent so you can monitor and revise
Software vs spreadsheet	Prefer software when hard to remember recording + prefer card purchases
Fix problems	Monitor to identify → gradual revision to fix

Envelope method: Put budgeted cash for each expense category into labeled envelopes; spend only that category's cash.

Borrowing / cash position

- Need to borrow when: **beginning cash + net cash < minimum cash balance** required for the month
- Cash budgets help lenders trust firm can operate within borrowing limits and repay
- Knowing cash position: **Businesses** → when short-term loans needed; **Individuals** → progress toward personal goals (individuals usually shouldn't need ST loans)
- Purpose of monthly cash budget: control inflows/outflows to balance income with expenditures and savings

W&H case hooks

- Labor costs = **cash disbursement**
- After budget: **monitor actual cash flows and revise if needed**
- Useful for loan: shows creditors firm can support operations within borrowing limits

B. Financial Forecasting (Longer-term)

Purpose / goal

Idea	Remember
Role	Supplements history with proposed investments/changes for better foresight
Goal	Understand implications of today's decisions on tomorrow's performance
Key question answered	How much financing will the firm need in the future?
Horizon contrast	Cash budget ≤ ~1 year; forecasting often 2–5 years (unit summary)
Long-term forecasts used for	Investment and financing decisions

Spontaneous vs discretionary accounts

Type	Meaning	Examples
Spontaneous	Vary naturally/proportionally with sales	COGS, cash, AR, inventory, accounts payable
Discretionary (non-spontaneous)	Do not vary automatically with sales; management decides	Notes payable, long-term liabilities/debt , common stock
Special (neither purely)	Fixed assets and retained earnings need special treatment	

If sales grow 10%/15% → increase spontaneous accounts (cash, AR, inventory, COGS) similarly; leave notes payable / LT liabilities constant unless management decides otherwise.

Percent of sales & DFN

Concept	Remember
Pro forma statements	Projected future financial statements
Percent of sales method	Sales forecast + historical relationships → pro formas
When DFN determined	After pro forma statements forecasted with % of sales
DFN	Additional financing needed given expected growth
DFN formula idea	Projected assets – projected liabilities – projected owners' equity
What DFN tells	Total funding management must obtain through discretionary financing sources
Negative DFN	Firm will have enough financing to fund projected sales (no extra financing needed)
Estimating BS changes with sales	Forecasting spontaneous accounts

Ways to decrease DFN

- **Increase plowback / retention** (↑ projected OE)
- **Increase net margin** (↑ projected OE)
- First place to check capacity constraints: **Fixed assets**

Do **not** decrease DFN by decreasing retention / increasing dividend payout / decreasing net margin.

Sustainable growth rate (SGR)

Idea	Remember
SGR	Growth rate firm can grow while maintaining present financial ratios without issuing new equity
Components	Profitability, asset use efficiency, capital structure, dividend policy
Also described	$ROE \times \text{plowback}$; max growth keeping ratios constant without new equity
↑ SGR	Decrease dividend payout (↑ retention)
Avoid new stock but grow fast as possible → look at SGR	

Fixed assets / capacity / lumpy assets

Idea	Remember
Why % of sales for FA unrealistic	Assets are lumpy — buy whole factory/equipment, not a fraction
Grow FA at capacity	Invest a substantial lump sum at once to increase capacity
Sales capacity equation	Shows how much room to grow without additional FA investment (actual sales / % of capacity)

Practice — Unit 5 (key Qs with answers)

Cash budgeting

1. Variances on cash budget indicate → managers/divisions **not meeting targets**
2. Cash budget forecast horizon → **1 month to 1 year**
3. Three budgeting principles (subset) → keep records; develop savings/income/expense strategies; method that meets needs
4. Three main uses → forecast need, performance evaluation, corrective action
5. Order to create cash budget → receipts → disbursements → create budget
6. Better Christmas goal → **specific and measurable** (\$50/month)
7. Mortgage fixed because → **same amount each month**
8. More efficient budget → compare budgeted vs actual, **revise**
9. Personal cash receipt example → **graduation gift \$100**
10. Business disbursement example → **rent check paid/cashed**
11. Business cash budget includes → receipts, disbursements, **borrowing**
12. Business disbursements examples → raw materials, rent, admin, interest, selling expenses
13. Sales ≠ receipts because → sales include **credit sales**
14. Why track cash flows → know where money goes; monitor & revise
15. Prefer software tracking when → forget to record + prefer card
16. Purpose of monitoring → actual vs goals; when correction needed
17. Identify & fix problems → **monitor** identify; **gradual revision** fix

18. Labor costs in budget → **cash disbursement**
19. After developing budget → **monitor and revise**
20. Useful for bank loan because → builds lender trust / shows can repay within limits
21. Cash position benefit: business vs individual → ST loans needed vs progress to personal goals
22. Tracking vs monitoring → monitoring uses tracking record to evaluate vs target/patterns

Unit 5 Test Form B highlights

1. Short-term cash sufficiency forecast → **Cash budget**
2. Purpose of monthly cash budget → control inflows/outflows; balance income, expenditures, savings
3. Need to borrow when → beg cash + net cash < **minimum cash balance**
4. Envelope method → cash in labeled envelopes by category
5. Budget changes → prioritize; implement **gradually one by one**
6. Cash/AR/inventory ↑ with sales % → **Spontaneous accounts**
7. Negative DFN → firm has **enough financing** for projected sales
8. Spontaneous account → **Accounts payable**
9. SGR useful because → max growth maintaining ratios **without new equity**
10. Buy whole factory for capacity → fixed assets are **lumpy**

Unit 5 Test Form A highlights

1. Jerry budgeting principles → keep records; understand savings/expenses/income; eliminate consumer debt
2. Before personal budget determine → **income, expenses, savings**
3. Sales \$100k, cash \$90k → not all sales on cash (credit)
4. Effective budget → **track, monitor, revise**
5. Envelope method → withdraw/set cash per category spending limit
6. Long-term forecasts used for → **investment and financing decisions**
7. Don't vary with sales; management discretion → **Non-spontaneous / discretionary**
8. Discretionary account → **Notes payable**
9. Grow without new equity → **Sustainable growth rate**
10. FA increase as lump sum because → must purchase **entire** asset, not just portion needed

Forecasting / DFN / SGR quizzes

1. Forecasting role → history + proposals for accurate foresight
2. Goal of forecasting → today's decisions → tomorrow's performance
3. Question answered → how much **financing needed in future**
4. Helps decision-making → impacts of today's actions on future
5. Fundamental purpose → estimate how cost/sales changes impact future CFs & financing needs
6. Spontaneous accounts → vary naturally with sales
7. Account type that changes with sales growth → spontaneous
8. Sales +10% might also +10% → **COGS**
9. When can DFN be determined → after pro formas via % of sales
10. Decreases DFN → **increasing plowback ratio**
11. % of sales; leave constant → **notes payable** (discretionary)
12. Estimating BS changes with sales → forecasting **spontaneous accounts**
13. SGR definition → grow maintaining ratios **without new equity**

14. Reduce DFN → **increase net margin**
 15. Increases SGR → **decreasing dividend payout**
 16. Check first for capacity/DFN → **fixed assets**
 17. Grow FA at capacity → invest substantial amount at once
 18. Sales capacity tells → room to grow without more FA
 19. DFN definition → additional financing needed given expected growth
 20. LT liabilities held constant → **discretionary account**
 21. DFN tells → funding to obtain via discretionary sources
 22. Grow without issuing stock → **SGR**
 23. Major purpose of forecasting → how decisions impact future growth
-

Quick Memory Hooks (Unit 5)

- Cash budget = **short-term** (1 mo–1 yr); forecast need + evaluate + correct
- Personal principles: know yourself; savings/income/expenses; strategies; records; right method; cut consumer debt
- Budget order: **receipts** → **disbursements** → **budget**
- Track → Monitor → Revise (gradually)
- Sales ≠ cash (credit); spontaneous ↑ with sales; discretionary = management (NP, LT debt)
- DFN after pro formas; $DFN = \text{proj } A - L - OE$; negative DFN = enough financing
- ↓ DFN: ↑ plowback, ↑ net margin; check **fixed assets** capacity first
- SGR = grow without **new equity**; ↑ SGR by ↓ dividends
- Fixed assets = **lumpy**; sales capacity = growth room without new FA

Unit 6 — Capital Budgeting, Valuation & Project Cash Flows (OA-heavy)

Ideal evaluation / why TVM & opportunity cost matter

Idea	Remember
Ideal method should consider	All relevant cash flows — otherwise you don't know how the project enhances firm value
Investment adds value when	PV of benefits > PV of costs
Opportunity cost (TVM)	What you forgo by tying money/time in an investment vs other options
Timing of CFs matters because	Related to opportunity cost — compare CFs to other projects; earlier CFs better (can reinvest)
Same total \$600k inflows, different timing	Prefer the project that receives cash most quickly (highest NPV via TVM) — e.g., Project D in the quiz
Non-cash opportunity cost example	Study instead of movies → opportunity cost is going to the movies
Opportunity cost definition	Loss of ability to use an asset for the next best project once invested elsewhere

NPV (Net Present Value)

Idea	Remember
Positive NPV	Project increases shareholder wealth / adds dollar value to firm → accept
Advantage	Calculates dollar value added to the firm by the project
Disadvantage	Not effective to compare projects of different sizes
Hardest / most important input	Cost of capital / required rate — inaccurate estimate → reject good projects or accept bad ones
Mutually exclusive projects	Use NPV (pick the one that adds most value) — e.g., among 4 projects, choose highest NPV (Project 2 : IRR 11%, NPV \$67k beats higher IRRs with lower NPV)
NPV = 0	Means IRR has been reached

IRR (Internal Rate of Return)

Idea	Remember
Definition	Rate that makes NPV = \$0
Why use for non-finance friend	Easy to interpret / communicate investment potential
Accept rule	Accept if IRR > hurdle rate / cost of capital (Bobby's Books: hurdle 9%, IRR 11% → yes)
Unconventional CFs problem	Multiple sign changes → multiple IRRs ; can't tell which is correct → use NPV
Mutually exclusive	IRR alone can mislead; prefer NPV for value added
Limitation (module)	Don't use for multiple sign changes or mutually exclusive ranking alone

PI (Profitability Index)

Idea	Remember
What it is	Presented as a ratio — discounted benefits / discounted costs
How it aids NPV	Scales different-sized projects so returns are comparable; gives idea of return generated
Accept rule	Accept if PI > 1 ; PI 0.8 → reject (inflows 20% short of covering investment)
Capital rationing / not mutually exclusive but limited budget	Rank/accept by highest PI first until capital used (Alphabet Co.; Betsy's Wigs: 1.35, 1.27, 1.02 → Projects 3, 2, 1)
Adjustment that can make PI < 1	Higher cost of capital decreases PI → may reject
Mutually exclusive	PI not the primary pick method (use NPV for mutually exclusive)

Decision cheat sheet: - Mutually exclusive → **NPV**

- Capital constraints / rank within budget → **PI**

- Communicate % return / simple story → **IRR** (if conventional CFs)

- Unconventional CFs → **NPV**

Why capital / returns / debt

Idea	Remember
Why firms seek capital investment	Purchase long-term assets for future growth
Capital budgeting	Evaluating & planning purchases of long-term assets
Why always a cost to bring funds in	Must compensate investors for risk of investing
Risk–return	Higher risk → higher required return
Debt benefit	Interest is paid before taxes (tax shield)
Financing decision	Proportions of debt and equity to fund assets/operations
Investor view of returns	Amount they require to lend/invest in the corporation's securities

Bonds & stocks (valuation basics)

Idea	Remember
Why raise via stocks vs bonds	Stocks do not require firm to repay par value to investors
Bond price < par (\$973 vs \$1000)	Discount bond (YTM > coupon)
Preferred stock vs bonds	Company can skip preferred payments (not bonds — bond skip = default); must eventually pay dividends in arrears before common
Bond valuation method	PV function in Excel (rate, nper, pmt, FV)
Common stock key assumption	Growth rate stays the same forever (constant growth / Gordon model)
Preferred stock valuation	Perpetuity model
YTM ↑ → bond price	Falls ; YTM ↓ (8% → 7%) → price increases (inverse)
Bonds	Fixed-income; fixed interest payments
Common stock	Variable dividends + voting rights
Preferred	Hybrid; fixed payments; generally no voting ; paid before common
CAPM (module)	Risk–return for an asset; way to price risk

Incremental / incidental / cannibalization

Concept	Remember
Incremental cash flows	Any additional cash flows in or out created by accepting a project
Allocated overhead	General firm cost, not a direct result of the project → irrelevant to analysis
Incidental vs non-incremental	Incidental = indirect but must be included ; non-incremental excluded
Cannibalization / cost of cannibalization	Reduction in sales of a company's own products due to introducing another similar product
Sunk costs	Already incurred whether you do the project or not → irrelevant (do not include)
Sunk examples	Market study already paid; consulting spent before project start
Not sunk	Required training; shipping new machine; forgone land sale offer

Beckingham Sports: \$400k study + \$600k consulting = **sunk** (ignore). Land cost \$2M but offer of **\$3M** to buy the land → relevant = **\$3M opportunity cost**.

Extra Unit 6 locks (Assessment Sec 1–2 — study these hard)

Idea	Remember
NPV = 0	Earns exactly required return; neither adds nor takes value
TVM valuation methods	Both NPV and IRR
Mutually exclusive	Use NPV (not PI for \$ value)
Prefer bonds over stocks	Bonds do not require giving up ownership/control
Bond like an annuity	Constant coupon each period + maturity date
Why cost of capital matters	Project CFs may be uncertain
Differently timed CFs	Must determine opportunity cost
Debt financing disadvantage	Does not achieve optimal capital structure alone — need debt + equity mix
NPV most reliable because	TVM + \$ value added + takes risk into account
Results as percentages	Used IRR
PI = 1	IRR = cost of capital (break-even)
Price = par	Par bond
Preferred = perpetuity	Fixed payment forever
Why TVM in evaluation	\$ today $\neq$ same \$ in 10 years
Clothes vs electric bill	Opportunity cost = electricity turned off
Seemingly identical projects	Different inherent risks → different cost of capital
Sunk cost example	Consulting spent 3 months prior to project

Traps (from missed Qs): - Bond $\neq$ constant growth forever (common stock); bond = coupons to maturity

- Cost of capital reason $\neq$ “reinvest earlier” (TVM) → **uncertainty**
- Don’t compare differently timed projects on benefits only → **opportunity cost**
- Debt disadvantage $\neq$ “can’t raise equity” → need **mix**, not all debt
- Land purchase may be sunk; **forgone \$3M sale** is the opportunity cost
- PI = 1 $\neq$ benefit outweighs cost by initial amount (that $\approx$ PI of 2)
- % answers → IRR; \$ answers → NPV

Practice — Unit 6 (correct answers)

NPV / IRR / PI basics

1. Project increases shareholder wealth → **NPV positive**
2. Important but hard to estimate in NPV → **Cost of capital**
3. Advantage of NPV → **Dollar value added to the firm**
4. Disadvantage of NPV → **Not effective to compare projects of different sizes**

5. Explain TVM-naive friend why invest → **IRR** (easy to interpret)
6. NPV when IRR reached → **\$0**
7. Unconventional CFs → multiple IRRs → use **NPV**
8. Hurdle 9%, IRR 11% → **Accept** (IRR > cost of capital)
9. Method presented as a ratio → **PI**
10. How PI aids NPV → idea of **return**; scales different sizes
11. PI 0.8 → **Reject** (short of covering investment)
12. Adjustment causing reject after PI=1 → **Higher cost of capital**

Securities / capital budgeting definition

1. Prefer stocks over bonds for capital because → no need to repay **par**
2. Bond bought below par → **Discount bond**
3. Preferred vs bonds → can **skip** preferred payments (not bond payments)
4. Why seek capital investment → buy **long-term assets** for growth
5. Bond value calculation → **Excel PV**
6. Common stock evaluation key assumption → **constant growth forever**
7. Process for LT asset purchases → **Capital budgeting**

Module / ranking applications

1. Accurate required rate important because → wrong rate → reject good / accept bad projects
2. One project only; maximize value → choose **highest NPV** (Project 2)
3. Not mutually exclusive; rank for most value → **highest PI first** (3, 2, 1)
4. YTM 8% → 7% → bond price **increases**
5. Cost \$100k, NPV \$5k → **Accept** (positive NPV)

Ideal method / mutually exclusive / rationing

1. Consider all relevant CFs because → know how project enhances firm value
2. Mutually exclusive → use **NPV**
3. Limited capital, maximize value created → projects with **highest PIs** until capital used

Timing / opportunity cost / financing

1. Opportunity cost in TVM → forgo other options due to time scope of investment
2. Timing of CFs important because → opportunity cost vs other projects
3. Four projects same total inflows → choose one getting cash **quickest** (Project D)
4. Always a cost to bring funds → compensate investors for **risk**
5. Risk and return → higher risk → higher required return
6. More debt can benefit → interest paid **before taxes**
7. Financing decision example → mix of **debt and equity**

Incremental CFs

1. Incremental CFs → additional CFs created by accepting project
2. Allocated overhead → **not** direct result; irrelevant

3. Incidental vs non-incremental → incidental **included**; non-incremental not
4. Cannibalization → new product steals sales from **your other product**
5. Limited budget, not mutually exclusive → prioritize with **PI**
6. Opportunity cost not cash flows → study vs movies
7. Investment adds value when → **PV benefits > PV costs**
8. How purchasers view returns → amount they require to lend/invest
9. Opportunity cost → can't use asset for next best project once invested elsewhere

Assessment Section 1–2 (new)

1. NPV = 0 → earns **exactly** the required rate
2. TVM evaluation — more than one way? → **Yes: NPV and IRR**
3. Mutually exclusive → **NPV**
4. Prefer bonds over stocks → don't give up **ownership**
5. Bond like annuity → constant coupon + maturity
6. Why cost of capital → CFs may be **uncertain**
7. Differently timed CFs need → **Opportunity cost**
8. Debt disadvantage → not optimal structure alone (need mix)
9. Sunk costs → **Irrelevant**
10. Beckingham relevant item → **\$3M land offer** (opportunity cost)
11. NPV most reliable → TVM + \$ value + risk
12. Results all % → **IRR**
13. $PI = 1$ → $IRR = \text{cost of capital}$
14. Buy bond at \$1000 par → **Par bond**
15. Preferred like perpetuity → fixed amount **forever**
16. Why TVM → \$ today $\neq$ \$ in 10 years
17. Clothes not bill → opportunity cost = power shut off
18. Seemingly same projects → different **inherent risks**
19. Own-product sales drop → **Cost of cannibalization**
20. Sunk example → consulting **before** project start

Quick Memory Hooks (Unit 6)

- +NPV = add wealth; NPV=0 = earn exactly required return; hard part = **cost of capital**
- NPV = \$ value; weak for different sizes → **PI** under capital constraints
- Mutually exclusive → **NPV**; % story → **IRR**; $IRR > \text{hurdle}$ → accept
- Weird CF signs → multiple IRRs → NPV
- $PI > 1$ accept; $PI = 1$ → $IRR = r$; raise r → PI falls
- Discount / par / premium bonds; YTM ↑ price ↓
- Stocks: no repay par; Bonds: keep **ownership**
- Bond $\approx$ annuity; preferred $\approx$ perpetuity; common = constant growth
- Include incremental + incidental (cannibalization); ignore overhead & **sunk costs**
- Forgone land sale = opportunity cost; prior consulting = sunk
- Debt tax shield BUT optimal = **debt + equity mix**
- Earlier cash better; differently timed → opportunity cost

- Answers in % → IRR; answers in \$ → NPV

Material Log

Source	Status
Unit 2 (all prior)	Captured
Unit 3 interest / TVM / Excel / returns	Captured
Risk lessons + quizzes	Captured
Ratio analysis + DuPont	Captured
Unit 5 cash budgeting + forecasting (DFN/SGR) + Forms A/B	Captured
Unit 6 NPV / IRR / PI + securities + incremental CFs	Captured
Unit 6 Assessment Section 1 & 2 (sunk costs, Beckingham, PI=1, traps)	Captured
Reddit OA tips	Strategy only
Quizlet	ON HOLD

(Add more units/forms/transcripts as provided.)